

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-ac1dd26a017831d49.jsonl`  
- SHA-256: `220327cbab784adc999d0bc9d4bf4e43b99ca9e872eb6f447efe773456c44692`  
- Source modified: `2026-06-06T17:31:01+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:30:14.122Z)

Source Extractor

Research question: "Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ? especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio)."

Fetch and extract key claims from this source:

URL: https://arxiv.org/abs/2411.08466

Title: Weakly-Supervised Temporal Action Localization Guided by Multimodal LLMs (MLLM4WTAL, arXiv 2411.08466)

Found via: broad/primary search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-06T17:30:17.367Z)

I'll fetch the source and extract key claims.

3. user (2026-06-06T17:30:29.411Z)

Paper Summary

Title: Weakly Supervised Temporal Action Localization via Dual-Prior Collaborative Learning Guided by Multimodal Large Language Models

Authors: Quan Zhang, Jinwei Fang, Rui Yuan, Xi Tang, Yuxin Qi, Ke Zhang, Chun Yuan

Submission Date: November 13, 2024 (v1); June 9, 2025 (v2, accepted to CVPR)

Abstract: The paper proposes MLLM4WTAL, a learning paradigm that leverages multimodal LLMs to enhance weakly-supervised temporal action localization.

Method Overview

How MLLMs are Used:

The approach harnesses MLLMs to provide "temporal action key semantics and complete semantic priors" for conventional WTAL methods. The system integrates two modules:

- **Key Semantic Matching (KSM):** Matches key semantic information from MLLMs
- **Complete Semantic Reconstruction (CSR):** Reconstructs complete semantic priors

MLLM Role (Training vs. Inference):

The abstract states the paradigm "facilitates the enhancement of WTAL by leveraging MLLM guidance," suggesting MLLMs function as "teachers during training" rather than inference components. The text emphasizes that "traditional models in video comprehension tasks" remain important, implying the final WTAL model operates independently.

Supervision/Labels Required:

The method addresses "weakly-supervised" settings, meaning only video-level action labels are needed (not frame-level annotations).

Performance Claims:

The abstract does not specify numerical results on THUMOS14 or ActivityNet. It only states experiments validate "efficacy of our proposed approach in augmenting the performance of various heterogeneous WTAL models."

****Key Claim:**** MLLMs address "incomplete and over-complete outcomes common in WTAL methods."

4. assistant (2026-06-06T17:30:32.029Z)

Let me get the exact abstract text and any specific quotes since the first fetch paraphrased some claims.

5. user (2026-06-06T17:30:39.410Z)

Paper Information

****Full Abstract (verbatim):****

"Recent breakthroughs in Multimodal Large Language Models (MLLMs) have gained significant recognition within the deep learning community, where the fusion of the Video Foundation Models (VFM) and Large Language Models (LLMs) has proven instrumental in constructing robust video understanding systems, effectively surmounting constraints associated with predefined visual tasks. These sophisticated MLLMs exhibit remarkable proficiency in comprehending videos, swiftly attaining unprecedented performance levels across diverse benchmarks. However, their operation demands substantial memory and computational resources, underscoring the continued importance of traditional models in video comprehension tasks. In this paper, we introduce a novel learning paradigm termed MLLM4WTAL. This paradigm harnesses the potential of MLLM to offer temporal action key semantics and complete semantic priors for conventional Weakly-supervised Temporal Action Localization (WTAL) methods. MLLM4WTAL facilitates the enhancement of WTAL by leveraging MLLM guidance. It achieves this by integrating two distinct modules: Key Semantic Matching (KSM) and Complete Semantic Reconstruction (CSR). These modules work in tandem to effectively address prevalent issues like incomplete and over-complete outcomes common in WTAL methods. Rigorous experiments are conducted to validate the efficacy of our proposed approach in augmenting the performance of various heterogeneous WTAL models."

****Subject/Category:**** cs.CV (Computer Vision and Pattern Recognition)

****Comments:**** Accepted to CVPR

****Submission Date Line:**** [Submitted on 13 Nov 2024 (v1), last revised 9 Jun 2025 (this version, v2)]

6. assistant (2026-06-06T17:30:44.816Z)

I now have the verbatim abstract and metadata. Let me assess and extract claims.

This is a primary research paper (arXiv preprint, accepted to CVPR 2025) from authors at Tsinghua University and collaborators. It is directly relevant to the "temporal action localization (TAL) deep models" and "VLM teacher?student distillation" families in the research question. The key insight bearing on the user's context: MLLMs are too compute-heavy to run at inference, so they are used only as guidance during training while a lightweight conventional model runs at inference ? exactly the teacher?student pattern the user is evaluating. However, the abstract gives no numeric performance figures and isn't badminton-specific.

7. user (2026-06-06T17:30:55.632Z)

Structured output provided successfully

8. assistant (2026-06-06T17:31:01.294Z)

Extracted and returned. Summary: arXiv:2411.08466 (MLLM4WTAL, CVPR 2025, Tsinghua et al.) is a primary research source directly relevant to the TAL-deep-model and VLM-teacher families. Its central bearing on the research question is that it validates the teacher-only architecture the user is already pursuing ? MLLMs are too heavy for inference ("their operation demands substantial memory and computational resources"), so they are used only as training-time guidance/priors while a lightweight conventional model does the actual localization at runtime. It also addresses the same failure modes the user faces (incomplete/over-complete action boundaries, analogous to WASB recall and boundary misalignment) and works in a weakly-

supervised, video-level-label-only regime. Caveat: the abstract reports no numeric accuracy figures and is generic action localization, not badminton-specific.